

Motion & Forces in a Gravitational Field

- Set 1
- 2 a) 18 m N 56° E
 - 2 b) 22 m N 63° W
 - 3 294 N
 - 4 3.8 m s^{-1} at 23° to rip
 - 5 a) 42°
 - 5 b) 3.4 m s^{-1} at 53° to the bank
 - 5 c) 30 m
 - 6 a) 9.2 km
 - 6 b) N 41° W
 - 7 5.5 m s^{-1} away from the player
 - 8 47 m s^{-1} toward the opponent
 - 9 32 m s^{-1} at 51° to the final velocity
 - 10 212 N in the forward direction
 - 11 42 m s^{-1} at 45° to both initial and final velocities
 - 12 a) Position 1: 427 N toward the Earth
 - 12 b) Position 2: 359 N at 2.10° to the line joining the Earth and the asteroid
 - 13 43 m s^{-1}
 - 14 1.78 m s^{-1} N 38.2° E
 - 15 a) $1.17 \times 10^3 \text{ N}$
 - 15 b) 125 N
 - 16 14 m s^{-1} , 20 m s^{-1}
 - 17 4.14 m s^{-2} down the slope
 - 18 2.5 s
 - 19 177 N perpendicular to, and toward, the path of the boat
 - 20 a) 3.5 s
 - 20 b) 30 m
 - 21 b) 32.6°
 - 22 716 N at 2.7° to the left of the boat's path
- Set 2
- 6 5 points
 - 7 3.48 m
 - 9 1.04 s
 - 10 1.16 m
 - 11 a) 2.1 s
 - 11 b) 31 m
 - 12 78.8 m
 - 13 0.81 m
 - 14 285 m
 - 15 a) vertical = 9.0 m s^{-1} upward; horizontal = 9.1 m s^{-1}
 - 15 b) 6.5 m
 - 15 c) 18.9 m
 - 16 a) vertical = 29.0 m s^{-1} ; horizontal = 7.76 m s^{-1}
 - 16 b) No
 - 16 c) 29.0 m s^{-1} to the right
 - 16 d) Assuming she hits the ramp with her foot already fully down, then at point A; Ignoring air resistance, the speed at point A should equal the speed at point E
 - 16 e) 27.3 m s^{-1}
 - 17 c) Yes, by 9.2 m
 - 18 35.6 m
 - 19 8.10 m s^{-1}
 - 20 a) no
 - 20 b) yes

- Set 3
- 5 0.82 m s^{-2} toward the centre of the circle
 - 6 61.9 N
 - 7 a) 1.54 m s^{-2}
 - 7 b) 17.5 N
 - 8 17.6°
 - 9 a) 691 N
 - 9 b) 3.55 s
 - 10 a) yes
 - 10 b) 3.09 kN
 - 10 c) 15.3°
 - 11 15.8 m s^{-1}
 - 12 a) 37.8 m s^{-1}
 - 12 b) $1.90 \times 10^4 \text{ m s}^{-2}$
 - 12 c) 17.7 kHz
 - 13 b) $2.59 \times 10^{-16} \text{ N}$
 - 13 d) $5.53 \times 10^8 \text{ m}$
 - 13 e) it would drop 30.6 m
 - 16 a) 88.5 m s^{-1}
 - b) 198 m s^{-1}
 - 17 87.4 m s^{-1}
 - 18 a) 180 N upward (b) 20.4 N downward
 - 19 a) 20 m
 - 20 b) 4.32 m s^{-1}
 - c) $2.00 \times 10^3 \text{ N}$
 - 21 b) 5.94 m s^{-1}
 - c) at top, 0; at bottom, 1176 N upward
 - d) at top, 441 N upward; at bottom, 735 N upward
 - 22 a) at A: 8.30 m s^{-1} ; at B: 12.1 m s^{-1}
 - b) at A: 61.6 N ; at B: 208 N
- Set 4
- 6 $6 \times 10^{24} \text{ kg}$
 - 7 $1.72 \times 10^{-6} \text{ N}$
 - 8 a) $2.64 \times 10^6 \text{ m}$
 - 8 b) 8.16 m s^{-2} toward the Earth
 - 8 c) $7.55 \times 10^3 \text{ m s}^{-1}$
 - 9 $3.80 \times 10^8 \text{ m}$
 - 11 b) $2.38 \times 10^{20} \text{ N}$ toward the Sun
 - 18 $5.74 \times 10^3 \text{ s}$ (1.59 hours)
 - 19 a) $1.37 \times 10^4 \text{ m s}^{-1}$
 - 19 b) $1.90 \times 10^{27} \text{ kg}$
 - 20 $5.97 \times 10^{24} \text{ kg}$
 - 21 $3.59 \times 10^7 \text{ m}$
 - 22 $3.05 \times (\text{radius of Moon's orbit around Earth})$
 - 23 a) $2.07 \times 10^{22} \text{ N}$; $9.20 \times 10^{21} \text{ N}$
 - 23 b) $5.37 \times 10^4 \text{ m s}^{-1}$; $4.39 \times 10^4 \text{ m s}^{-1}$
- Set 5
- 2 120 N m
 - 3 220 N
 - 6 b) 18 kg
 - 8 a) 94 N
 - 8 b) 447 N
 - 9 46 kg
 - 10 a) 630 N
 - 10 b) 0.75 m toward Q
 - 11 1.5 m from the front wheels
 - 12 a) 18 kN
 - 12 b) $0.375 \times \text{length of log}$ from the heavier end

- 14 a) 383 mm from the balcony
 15 $1.86 \times 10^5 \text{ N}$; $1.95 \times 10^5 \text{ N}$
 20 7.76 kg; 2.57 m from the T_1 end.
 21 60 cm
 23 a) 137 N
 b) lower hinge 172 N toward the door at 36.9° above the horizontal
 upper hinge 172 N toward the wall at 36.9° above the horizontal
 24 a) $1.95 \times 10^3 \text{ N}$
 b) vertical = 109 N upward, horizontal = $1.83 \times 10^3 \text{ N}$ out from wall
 c) $1.84 \times 10^3 \text{ N}$ out from wall at 3.40° above the horizontal
 25 tension = $1.09 \times 10^3 \text{ N}$; force at hinge = 551 N out from the wall and 10.1° below the horizontal
 26 2.07 m from P
 27 a) 156 N
 b) vertical component = 147 N, horizontal component = 156 N
 c) 214 N to right and 43.3° below the horizontal
 28 a) $2.39 \times 10^5 \text{ N}$
 b) $2.97 \times 10^5 \text{ N}$
 29 a) 18.4 kg
 b) 144 N at 26.9° above the horizontal

Electricity and Magnetism

- Set 6 5 0.042 N m
 10 2.4 N
 12 0.075 N downward
 Set 7 1 0.57 V
 2 a) 2.3 A
 5 a) 0.21 mV
 5 b) $2.1 \times 10^{-4} \text{ Wb s}^{-1}$
 7 7.1 mV
 8 a) 39 mV
 8 d) 7.9 mA
 9 a) $8.0 \times 10^{-4} \text{ V}$
 11 225 turns
 12 a) 2.5 m s^{-1}
 12 b) $1.0 \times 10^{-4} \text{ N}$
 13 a) 83 mT
 Set 8 1 a) 60 kW
 1 b) 300 J
 1 c) 150Ω
 2 a) 14.6 m^2
 2 b) 1.25Ω
 3 a) 83 A
 3 b) 18Ω
 6 a) 0.0200 times
 6 b) 10 000 turns
 6 c) 0.0196 times
 7 a) 23Ω
 7 b) $1.3 \times 10^3 \text{ C}$
 8 a) 100 A
 10 a) 31.3 kW
 10 b) 0.31 kW
 12 a) 2.00 kW
 12 b) 0.125 m
 13 a) 6.25 kW
 15 35 km

Particles, Waves and Quanta

- Set 9
- 1 5 °C
 - 2 420 m
 - 3 82 m
 - 4 a) 50 mm
 - 4 b) 0.4 m s^{-1}
 - 4 c) 8.4 Hz
 - 5 a) 0.5 Hz
 - 5 b) 2 s
 - 6 a) 10 mm
 - 6 b) 8 ?s
 - 6 c) 125 kHz
 - 9 a) 1.3 m
 - 9 b) 17 mm
 - 9 c) 4.9 mm
 - 9 d) 3.4 km
 - 13 b) too short
 - 13 c) 0.29 s
 - 14 a) George
 - 14 b) Jane
 - 14 c) George
- Set 10
- 3 c) 34 Hz to 17 kHz
 - 5 a) 1.9 mm in air; 8.1 mm in water
 - 6 72.8 m
 - 7 a) 100
 - 7 b) 0.8°
 - 7 c) 10 scans
 - 12 708 Hz
 - 13 a) 512 m s^{-1}
 - 13 c) 84 Hz, 168 Hz, 262 Hz
 - 18 a) 0.64 m
 - 18 b) 328 m s^{-1}
 - 18 c) closed
 - 19 a) violin
 - 19 b) double bass
- Set 11
- 1 a) 3.0 GHz
 - 1 b) 10cm
 - 2 red: $4.41 \times 10^{14} \text{ Hz}$; $2.92 \times 10^{-19} \text{ J}$
orange: $5.17 \times 10^{14} \text{ Hz}$; $3.43 \times 10^{-19} \text{ J}$
green: $6.00 \times 10^{14} \text{ Hz}$; $3.98 \times 10^{-19} \text{ J}$
 - 3 0.001:1; 1000:1
 - 4 a) red
 - 4 b) $2.87 \times 10^{-19} \text{ J per photon}$
 - 4 c) $1.00 \times 10^5 \text{ W m}^{-2}$
 - 4 d) 100:1
 - 5 7.5×10^{25} photons per second
 - 6 5.1×10^{10} photons per second
 - 7 a) 420 m
 - 7 b) $4.3 \times 10^9 \text{ J}$
- Set 12
- 3 $E_1 = -13.6 \text{ eV}$
 $E_2 = -3.4 \text{ eV}$
 $E_3 = -1.5 \text{ eV}$
 $E_4 = -0.85 \text{ eV}$
 $E_5 = -0.54 \text{ eV}$

- 4 a) (i) 2.4×10^{15} Hz; UV
 (ii) 6.14×10^{14} Hz; visible
 (iii) 4.56×10^{14} Hz; visible
 (iv) 2.32×10^{14} Hz; IR
- 4 b) 4.58×10^{-19} J; E_5 to E_2
- 5 a) 45 keV
- 5 c) 30 keV; 40 keV
- 5 e) 2.0×10^{-16} J
- 7 a) 5.13 eV
- 7 b) 2.11 eV; E_2 to E_1
- 7 c) 0.002 eV
- 11 a) 60 keV (9.6×10^{-15} J)
- 11 b) 1.45×10^8 m s⁻¹
- 11 c) 2.07×10^{-11} m
- 12 12.4 kV
- 13 3.71×10^{-11} m
- 14 b) 4.14×10^{-11} m; 7.24×10^{18} Hz

Motion and Forces in Electric and Magnetic Fields

- Set 13 1 4.5×10^6 N C⁻¹
- 3 b) 2.94×10^7 m s⁻¹, deflected by 8.9°
- 3 e) 2.9×10^{-16} J
- 3 f) 6.0×10^4 V m⁻¹
- 5 a) 3.3×10^{-7} J
- 5 b) 66 V
- 6 100 V m⁻¹
- 7 a) 5.00 keV
- 7 b) 8.00×10^{-16} J
- 8 a) 10.0 keV
- 8 b) 1.6×10^{-15} J
- 9 a) 5.6×10^7 V m⁻¹
- 9 b) 2.4×10^{-15} J
- 10 b) 3.6×10^{-5} N
- 10 c) 10 mm; 0.090 V
- 13 3.91×10^5 m s⁻¹
- 14 b) 2500 eV (4.00×10^{-16} J)
- 14 c) 2.96×10^7 m s⁻¹
- 14 e) same
- 14 f) less
- 15 a) 4000 eV; 3.75×10^7 m s⁻¹
- 15 b) 2000 V m⁻¹; 3.20×10^{-16} N
- 16 b) 7.02×10^{15} m s⁻²
- Set 14 7 1.58×10^{-26} kg to 1.62×10^{-26} kg
- 8 a) 41.8 m
- 8 b) 6.28×10^{-3} s
- 9 b) 2.4×10^{-14} N
- 10 b) 6.3×10^7 C kg⁻¹; 4.8×10^7 C kg⁻¹
- Set 15 3 a) 1.10×10^5 V m⁻¹
- 3 b) Yes
- 4 c) They accelerate in a straight line at 1.76×10^{14} m s⁻²
- 5 a) 7.26×10^7 m s⁻¹
- 5 b) 1.76×10^{-4} m
- 5 c) 1.52×10^{-11} s

- 6 a) $7.19 \times 10^6 \text{ m s}^{-1}$
- 6 b) $2.29 \times 10^7 \text{ Hz}$; $4.38 \times 10^{-8} \text{ s}$
- 6 c) 270 kV
- 7 c) $1.00 \times 10^5 \text{ m s}^{-1}$
- 8 a) $3.75 \times 10^{13} \text{ m s}^{-2}$ at right angles to the field
- 8 c) $3.92 \times 10^5 \text{ N C}^{-1}$
- 9 a) $3.04 \times 10^{-6} \text{ m}$
- 9 b) 480 N C^{-1}
- 10 b) $2.87 \times 10^7 \text{ m s}^{-1}$; $5.75 \times 10^7 \text{ m s}^{-1}$